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Andean audiotactility: transcultural interfaces elucidating a divergent history of technology

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Abstract

This paper deals with the complex relationship between human beings, technical objects and the natural environment within a transdisciplinary framework at the intersection of human-computer interaction, philosophy, and materiality. It begins by examining the role of tactile coding systems in Andean fiber support, then draws upon diverse disciplines including art, computer science, philosophy of technology and media archaeology. Through the analysis of historical artifacts, contemporary artistic practices and philosophical frameworks, it aims to explore dynamic methodologies beyond the Western canon. The goal is to question entrenched power structures in a framework that foregrounds interconnectedness as a relational ontology inspired by Andean worldviews, embracing syncretic approaches so to reformulate technological narratives.

Keywords

technology; philosophy; media archaeology; transdisciplinarity; human-computer interaction; art; decoloniality; Andean traditions

Audiotactilidad andina: interfaces transculturales que aclaran una historia divergente de la tecnología

Resumen

Este documento trata de la compleja relación entre seres humanos, objetos técnicos y el entorno natural dentro de un marco transdisciplinar en la intersección de la interacción entre humanos y ordenadores, la filosofía y la materialidad. Comienza examinando el papel de los sistemas de codificación táctiles en soporte de fibra andina y, a continuación, recurre a diversas disciplinas, como el arte, la informática, la

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filosofía de la tecnología y la arqueología de los medios. A través del análisis de artefactos históricos, prácticas artísticas contemporáneas y marcos filosóficos, tiene como objetivo explorar metodologías dinámicas más allá del canon occidental. El objetivo es cuestionar las estructuras de poder arraigadas en un marco que antecede la interconectividad como una oncología relacional inspirada en las visiones andinas del mundo, adoptando enfoques sincréticos para reformular las narrativas tecnológicas.

Palabras clave

tecnología; filosofía; arqueología de los medios de comunicación; transdisciplinaridad; interacción humano-ordenador; arte; descolonialidad; tradiciones andinas

Introduction

My current research on textile technologies begins from a media archaeology perspective (Parikka 2023, 4), departing from the Khipu (Knill 2018), a textile technology of pre-Columbian origin, as an example of body-centric technology (Benford *et al.* 2017, 127). The study of khipus, an Andean notation system employing knots for information encoding, not only draws from anthropology and ethnology but incorporates the potential of new media as digital transcultural interfaces.

Media archaeology, a field that attempts to understand new and emerging media through close examination of their history and lineage, becomes key in examining this ancient fiber-based storage device as akin to writing, expanding the alternative history of computing. By studying the khipu, I explore an alternative information-encoding method embedded in a tradition where narratives are woven into textile fibers, relying on the human body for both encoding and decoding. The act of tying and untying knots, a characteristic of some of my sound art presentations, carries inherent performative qualities. This approach emphasizes the undocking of code from traditional text or numerical matrices, the former rooted in a unique lifeworld that shaped programming itself. This holistic knowledge beyond the audiovisual, often overlooked as primitive, offers a unique philosophical perspective that is nevertheless still dealt "just" as a "worldview" from indigenous thought systems. This knowledge, now known as traditional ecological knowledge (TEK) (Gann *et al.* 2019), is crucial in the face of the current climate crisis and provides a path to bring forward the natural world, but is still greatly overlooked by contemporary academia.

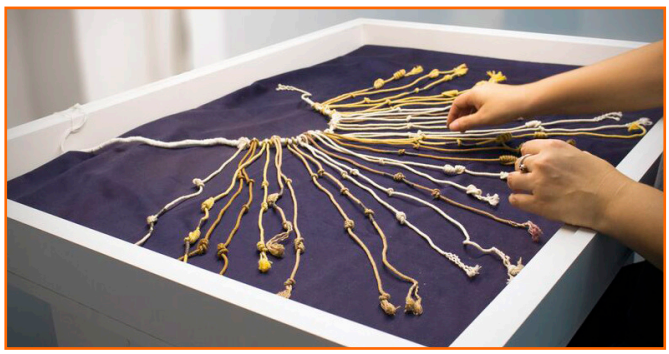


Figure 1. *The Hanap Pacha Khipu*, A Contemporary Khipu as sound controller, part of the Sonified Textiles. Source: photo by the author

As I grew up in Perú, the khipu was presented to me as the technology developed by my ancestors. Given this insider status, my research has inevitably adopted an autoethnographic approach, embracing my hybrid condition as a Peruvian mestiza, and acknowledging that the khipu is part of my identity as an Andean inhabitant, hence, stating that I design and narrate from such position. I also situate myself among a group of artists and creative technologists from South America who have been using the khipu as a source of inspiration and a tool for decolonial critique. Our collective practices, shaped by identity issues arising from our non-Western upbringing in environments heavily influenced by the West, have resulted in a strong yet mostly unrecognized presence of the khipu in artworks within the fine arts as well as in the creation of syncretic human-computer interfaces (Torres 2021) that mirror our hybrid (Flores 2014) identities, for example, a characteristic of the work of Patricia Cadavid and her Electronic_Khipu or Constanza Piña´s *Khipu Electrotextile Prehispanic Computer* (Cadavid 2020). Nevertheless, my investigation within textile controllers for sound underscores the significance of craft as *techné* (James 2019) and the impact of gesture in performing interactivity (Dixon 2007) while delving into how all involved agents are transformed using these interfaces, exploring changes in bodily gestures and the uniqueness of performances enabled by digital/electronic means. For a cross-examination of the mainstream *History of Technology*, I consider the theoretical aspect of this research to start within postphenomenology yet progressively shifting toward a more decolonial approach.

1. Tactile coding and embodied interaction

I have advanced the development of computational devices and interfaces that are employed in my own performances and sound art presentations, some initially conceived within the context of *mestizo* Latin American culture but developed within Sweden over the past decade. The research extends to exploring a cohort of Latin American artists at the forefront of interface design, integrating indigenous techniques inspired by pre-Hispanic cultural developments, with a specific focus on Pan-Andean worldviews, harmonizing technology with the natural world.

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Tactile coding, as exemplified by the khipu, allows the reimagining of interface design in the digital age. Embodied interaction, a core principle of HCI (human-computer interaction), emphasizes the importance of considering the human body as an integral part of the interaction process. This notion of embodiment posits that our bodily experiences play a crucial role in shaping our encounters with technology, influencing how we perceive, interpret and interact with technological artifacts. Already in their seminal 1997 paper "Tangible Bits: Towards Seamless Interfaces between People, Bits and Atoms", (Ishii & Brygg 1997) Hiroshi Ishii and Ulmer Brygg described how users interact with bits, the fundamental units of digital information, by manipulating physical objects and architectural surfaces in their everyday environment, anticipating the contemporary notion of the Internet Of Things, in which digital capabilities are embedded into physical objects. Nevertheless, they describe a very linear history of technology in such a paper, enumerating technical objects found in the Peabody Harvard Museum, completely ignoring the khipus found in such place, missing that this integral approach was embedded in informational devices as these, what it is now known as body-centric technologies when reflecting the idiosyncrasy of its original designers (the body as part of the *world-as-natur*e). It is from this integral perspective from which I have been departing to design what I call textile controllers for sound, or simply, the "sonified textiles," fiber hand-crafted HCI devices that allow control of digital sound generation. The tangible aspect fits within the field of human-computer interaction and can be extrapolated towards the acknowledging of the current dual human existence unveiling in both physical and virtual realms, a concept that has become increasingly relevant in the digital age while acknowledging the role of interfaces in such scheme.

2. Towards a "fourth wave" of HCI?

Human-computer interaction researchers in the Global North commonly use concepts and theoretical frameworks from philosophy as lenses for interpreting and reflectingon technology and its use (Svanæs 2019) and, in this line, the phenomenological approach has been of preference within the area of interaction design, particularly within the "third wave" HCI that went beyond the first wave that was model-driven and rigidly nested in formal methods and systematic testing, and the second wave that started with considering situated action and context (Bødker 2015) to then pass to the emphasizing the terms and conditions of the interactive relationship prior to the relation of the human subject and its computational object (Gunkel 2018) albeit these "terms and conditions" seemingly keep an user-centric approach where social and/or economic aspects might play a role in the recognition of the environment where this "user" is placed at, but all that lies beyond this defined area seems to be largely omitted. According to Susanne Bødker, HCI theory has widened from its root in engineering research by the incorporation of cognitive science, sociological studies, emotional design as well as social participatory research, yet a more recent shift openly claims for decentering the human within HCI, a move now framed as *Post-Human HCI*. For Seda Özçetin and Heather Wiltse, for example, this move implies taking into consideration multiple actors, their intentions and capabilities, and consequently their interrelations in entire ecosystems making the contemporary connected things thus making the experience of interacting more complicated or challenging by asking users to "engage with complex issues that are not relevant to their own purposes or immediate use" (Özçetin & Wiltse 2023). The fourth wave should turn towards pressing issues such as tackling the climate emergency and growing inequality so as to take "political activism from the margins into the mainstream of HCI" (Ashby *et al*. 2019). Posthumanist HCI then encompasses a broad variety of different perspectives such as science and technology studies (STS), object-oriented ontology (OOO), new materialism, agential realism and postphenomenology (Nicenboim 2023), the latter becoming particularly relevant when presenting interactive interfaces framed as "transcultural." This is due to their multi-stable condition (Ihde 1999, 46-47), a characteristic of technological artifacts in that they are not one thing or the other, being able to belong to multiple contexts.

Postphenomenology delves into the nuanced ways in which technologies mediate human experiences and how it shapes its (cultural) practices, presenting an alternative notion that moves beyond instrumentalist conceptions of technology. It emphasizes instead the co-constitutive nature of human-technology relations (Verbeek 2001, 9-41) within the world. At the core of postphenomenological inquiry lies the concept of technological mediation, which harnesses the transformative role of technologies in shaping our human perceptions, interactions and modes of existence. This, in turn, challenges the simplistic dichotomy between humans and machines, highlighting the dynamic interplay and mutual reshaping between technological artifacts and human culture on a wider scale. Verbeek goes to the extreme to state that when a technological artifact is used, it facilitates people's involvement with reality co-shaping "how humans can be present in their world and their world for them" (Verbeek 2006). By examining how technologies mediate our engagement with the world, postphenomenology sheds light on the multifaceted ways in which technology *in-forms* our lived experiences, even beyond the socioeconomic or user-centred level.

In a known set of progressive graphic representations by Don Ihde, that depicts the relationship between humans, technology and the world, (Idhe 1990, 89) he nevertheless shows a clear distinction between the active roles of humans engaged with technology as potentially transforming a world depicted at a distance from the human itself, to which we are inevitably connected with but not as part of it, in such a way that, culture and nature, under this framework, are still two different things, yet both "shaped" by human technology. Thus, one could conclude that even if framed as *posthuman* (Rosenberger & Verbeek 2015) it still maintains the anthropocentric perspective from which the human-technology-world interactions are formulated, as it still occurs against the backdrop of nature, a world that we experience

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and reshape (technically/technologically) and in turn reshapes us, yet it is separated from us.

The key question that arises from this framework is: what if we no longer consider the natural world as merely a background to experience, model and control as domination? How would this shift in perspective prompt us to reconsider not only the role of technology but the possibilities of new technologies designed under this "new" paradigm? Would this factual posthuman framework change our relation (reincorporation) to nature through technology, and vice versa, when we are assumed as incarnated beings or *instances* of the natural world?

Actor-network theory (Latour 2005) has indeed opened a door for discussing the undoubted connections between the social and natural worlds which exist as constantly shifting networks of relationships. In fact, we increasingly realize that these relational ontologies are becoming more popular (i.e., hydrofeminism) (Niemanis 2012), yet, in a way, this can be taken as a healthy rediscovery of much older philosophical traditions.

It is at this point where my research connects with *variantology*, when the need for acknowledging other ways of knowing, of living and relating with nature (Whyte 2013) and the designing of technological objects and their relationship with the arts and the science-as-systematized-knowledge becomes imperative. I start by acknowledging the khipu as being brought to life again from the burial of dead media (Anonymous 2010), so to continue with the explicit harnessing of historical omissions that would allow us to rethink technology itself.

3. Other ways of knowing embracing hybrid, transcultural interfaces

Anthropologists have recognized the persistence of khipu's use beyond colonial times in the Andean region. Researchers such as my own great-uncle, Oscar Nuñez del Prado, or more recently, Frank Salomon and Sabine Hyland, have rescued modern khipus from apparent oblivion when acknowledging the use of khipus in their publications demonstrating a usage that has continued even beyond the 20th century. Nevertheless, not only the technology itself has changed, but its role in Andean society. For example, in the case of the community of San Andrés de Tupicocha (in the highlands of Lima, in the Huarochiri region) the khipus have become a symbol of modern civic status, pointing out that the current user or the one possessing it, has taken a particular role within the community (Salomon 2004). These khipus, commonly stored by the community *Ayllus* (extended families, now known as *Parcialidades*) are shown in public in certain communal civic ceremonies, placed on the inhabitant to be used momentarily, and in a symbolic way, indicating his new communal leadership position. Although mostly made in Republican times (although some seem to go back to colonial times), these khipus seem to maintain the Inca structure, although the method of encoding information according to the Incan method, or the ability to interpret them in that way, is lost or was lost in the mid-20th century, according to their own narration. Yet the obscure practice of "reading the future" in some of these khipus is maintained by one of these *Ayllus* to date, another example of the *afterlife* of the fiber devices.

Diverse communities in the Andean region have continued using khipu-like structures to record data, such as the Q´ero community in the Highlands of Cusco (Núñez del Prado 1950), while in other regions these have become syncretic and have merged with imported European traditions and Christian faiths. An example is the khipus of Cuspón, in Ancash, a region north of Lima. These khipus are particularly relevant to the concept of *Andean audiotactility* that I put forward in this paper, even if they do not have the base structure of the known Inca khipu; that is, they do not consist of a main cord with various pendant cords hanging from it.

The khipus of Cuspón consists of one single, long fiber rope, made from industrial wool or even synthetic materials, tied in knots the size of a human hand. Although these do not follow the main Inca (and Wari) configuration of a cord with smaller ones hanging from it, it maintains the Inca knot style of encoding information depending on how many turns each knot has throughout the unique rope. These types of khipus are called "funerary khipus," because they are connected, in a very direct way, to the recently deceased, as they are made for the person passing to the afterlife to be buried with them. Traditionally made by women, these were thought to protect the dead. In Ancash, there are stories and myths that narrate how the khipus spared the turning into what we would understand as "the living dead" (Hyland *et al.* 2021). That is why a khipu of this type is placed over the corpse, before being buried, intentionally encoding protecting songs or prayers in their knots, information that can be read aloud while touching each of the knots as if these were a Christian rosary of sorts. In fact, the connection with such Christian mnemotechnic device (the Christian rosary) is openly recognized by Cuspón inhabitants (Zubieta 2017, 247-254). This does not only mean that khipus are a type of *external memory* (Schönpflug 2001, 5192-5195) that continued to develop beyond the invasion of the Americas by being transformed into a transcultural object, but that we can recognize how the actual mapping happened: through similarities. In this case, rosaries are used to remember prayers through touch, thus related to human language, potentially being able to be spoken aloud or even sung. When one takes into account that the colonial chronicles wrote about pre-Columbian khipus storing complex narrations and songs as music encoded in their structure, one can understand that the actual mappings in between these two cultures happen where similarities could be found, as mentioned beforehand, in the design of the object itself: both were thought to be hand-held, the knots found in the cords were able to be seen as correspondent with the rosary's small spheres that allowed the user to remember how many times he or she had to pray, and which type of prayer, portability, etc.

While Continental philosophy has provided valuable insights into the nature of human-technology relations, it often falls short in addressing the complexities of diverse media as *cosmotechnics* (Hui

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2017), coloniality and its impact on technological development and value, particularly when attempting to address how our modern technology impacts the natural world (Hui´s cosmopolitics). To manage this transcultural gap in the case of Andean technologies, I turn to the work of Anibal Quijano, the prominent proponent of decoloniality, asserting that Modernity emerged from the colonization of the Americas (Quijano 2007). This serves as a foundational premise for the exploration of the relations of modern technology and coloniality, framing it within the "decolonial turn in critical technology" that acknowledges that decolonial struggles against data extraction, struggles over particular practices of technology and struggles over knowledge and rationality (Couldry & Ali Mejias, 2023) to which I would also add the struggle from the perspective of the historiography of technology.

To illustrate this point further, I would like to share an aspect of my ongoing work, which involves a performance piece related to live coding. At the center of this performance is a large khipu, with which I engage by tying knots while interacting with a computer equipped with an intelligent computer vision system (machine learning). Over time, I have trained this system to recognize specific knot patterns and respond in various sonic ways so I can code music relating audio and haptics in real time.



Figure 2. The live coding performance known as *Khipumancy* alongside Patricia Cadavid
Source: photo composite by the author from two screen captures (second one from her own YouTube video: https://youtu.be/yA4hR9DS82k?)

Initially, the system engaged in live coding based on the knot positions. However, it has since evolved to not only generate music in real time but also compose poetry in the Quechua language based on combinatorics of three-knot combinations (including 0, or no-knot) as words. During the performance, the knots I tie are detected by a computer vision "AI" system, triggering the generation of music and poetry through live coding. This performance is accompanied by a video presentation of the manifesto, titled the *Neokhipukmayoq manifesto*, which puts forward the possibility of a new technology branching from that other history of technology that was cut short by the European colonization of the Americas. The manifesto draws on Andean concepts of interconnectedness and reciprocity, and in such a framework, I propose a reimagined technology that challenges paradigms of domination and extractivism. This methodological framework is rooted in the Andean concept of *ayllu* and emphasizes the interconnectedness of all beings while seeking a sustainable relationship between humans and technology under the new/old paradigm that considers humans as part of nature, under the traditional Andean philosophy of indigenous origin, that would go beyond the assumed binary of nature vs. culture.

The Andean *ayllu*, the extended family as the nucleus of an ancestral social structure based on kinship and reciprocity that still exists, offers insights into alternative modes of organizing society beyond a strict divide with nature. The dominant scheme applied to the natural world, including the extraction of knowledge between the Americas and Europe, serves as a support for understanding the intertwined histories of technology and colonialism. So, through an examination of syncretism within Andean culture, I highlight the complex entanglements between indigenous knowledge systems and colonial power structures (Quijano 1997). Quijano's own *mestizo* identity underscores the hybridity of cultural identities in South America and serves as a lens through which to analyze the intersections of technology, culture and coloniality.

Building upon the historical connections between technology and coloniality, I interrogate the role of textiles in shaping the history of computing. From Ada Lovelace's pioneering work in algorithmic thinking to the use of core rope memory in the Apollo missions, textiles have played a fundamental role beyond just serving as a source of inspiration for the development of computing technologies. The foregrounding of the contributions of women weavers and the influence of textile traditions on early computing are challenging conventional narratives of technological innovation. My performance related to live coding and AI computer vision, known as *KhipuKoding* or *Khipumancy* when presented alongside Patricia Cadavid and her own MIDI controlled khipu, inserts itself in this reformulating of technological historiography, expanding it beyond the Eurocentric tradition.

4. Designing generative interfaces as a research method

Marisol de la Cadena refers to the interface as a method to acknowledge this in-betweenness as a space for generation. She states that "it can provide alter uses for AND – that of fusing what it separates: making the interface nature and humanity active, making an event of/at the interface." She further discussed the idea of "Not only" (being something as well as something else) and that this creates possibilities for interfaces and for emergences at these interfaces that can force habits of thoughts that are alter-native, as "they are alter to our native thought practices." (de la Cadena 2015).

I am developing a dynamic methodological framework for transdisciplinarity, when it does not involve working with groups of experts specializing in diverse disciplines, but rather, when the individual pursuing the research uses various media and delves into diverse lineages of knowledge until their limits become blurred or the disciplines entangle. Drawing on concepts from systems and network theory, but, of the concept of *Ayllu* that includes both individual (subjective) and collective (shared) experience, I propose a flexible approach that accommodates

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multiple disciplinary perspectives and temporalities. This framework emphasizes the fluidity of disciplinary boundaries when seen as overlapping, separating and joining again in a node-based spatiotemporal dynamic network structure of knowledge that is assumed to change over time.

My research involves a continued interrelation of theory with practice in a revolving manner, integrating insights from complex thinking as a transdisciplinary method (Morin 1992), so to answer the call for a fourth wave of HCI by incorporating perspectives from Anibal Quijano's concept of "coloniality of knowledge" so to "decolonize technology" but the conventional view of human society as well. The research's significance extends to advancing the philosophy of technology beyond the Western framework by looking at it through a hybrid lens: that of the mestizo experience beyond acculturation, the mestizo as interface, as "not only" in de la Cadena´s terminology, as *multistable*, assuming the variantology of the media that emerged during human history, that inevitably shaped both humans and the planet.

One of the aspects of my research is to produce the definition of a method in the making, which is what I have been applying to develop my own work in the arts and in technology, and its theoretical framework and that I would define as non-linear and multilayered. My goal with this tentative definition is to point out that my research method is of a dynamic nature and can be placed within a different predefined research category as discipline depending on the moment or the position it is seen from, yet assuming that this is a snapshot of an interconnected, ever-changing structure whose intersections usually end up in material form, either as text, as an artwork, or as a textile for human-computer interaction, an intersection that becomes generative interface.

5. A possible conclusion: the shift from "relational ontology" to *Awasqa kayku*

According to Gloria Schaab, relational ontology is the philosophical position that distinguishes subject from subject, subject from object, or object from object and instead focuses on its "mutual relation rather than in its substance" (Schaab 2013). Ontologically, substance refers to the essence or nature of a being. The concept of "substance" is commonly treated within philosophy as that of an object, or thing when this is contrasted with properties, attributes or events. To frame my research within a relational ontology could be a starting point to break free from the Western paradigm of thought, characterized by its tendency to dualism, based on a binary logic (true/false) and this has been reflected in such a division (culture vs. nature) as well as in the split of (human) knowledge into various disciplines. Hence, situating it specifically within transdisciplinarity, instead of, say, interdisciplinarity, which involves the collaboration across differences – in turn embracing such differences – underlines its integrated approach. Yet "relational ontologies" as such are also defined from hegemony. To transcend such tendency towards dualism and eurocentrism, embedded in the history of academia itself and by taking each of the textile controllers not only as cross-cultural, as generative interfaces, but also, being polysemic and syncretic (having multiple meanings and cultural references) I recognize that these so-called relational ontologies are nothing but new, instead shaping how knowledge was produced and disseminated in various ancestral cultures (i.e. Ubuntu philosophy, *Te Ao Māori*, Australian Aboriginal "Pattern Thinking", etc.) hence the need for a terminology that harnesses the lineage of Andean audiotactile technologies beyond the English language, as an alter-native.

At this point, I am framing my method as an ecology of meanings but also of aesthetics, processes and events: as the substance of these variant interfaces in between technology, science and art yet shifting the gaze not only to another type of philosophy, one that would not only originate in the "not only", in the in-between of Western and non-Western, that is, Andean syncretic worldviews, but that would emphasize on the actual links in between all matter and forces, going beyond organic compounds: I am calling this new framework *Awasqa kayku*, where *Awasqa* means both *weave* and *pattern* in Runa Simi (the Andean language known as Quechua), that, when placed together with the word kayku it could be translated as "we are woven" as well as "we are pattern". This is a relational ontology yet coming from another philosophical tradition while embracing Quechua terminology coined within the Westernized post-colonial techno-culture.

Avrina Jos, a researcher of Indian origin who writes from Europe thinking about the Global South (very much like myself) discusses this problem in her paper "Whose Emancipatory Politics? Toward a Post-colonial Technological Subject" where she starts by wondering how to create representational imaginaries of technology, something that attains my research profoundly, that would "bring to the table people whose struggles are barely recorded but on whose backs histories are written," that is, those bodies shaped by "terrible histories of partition, colonization, casteism, and untouchability" and how these could potentially have a very different relation to technology (both to its historicity and accessibility) or just come up with another type of technology imaginary altogether (Jos 2021). She concludes that the lack of a post-colonial perspective in these Western feminist imaginaries reproduces a very monolithic or Eurocentric knowledge, a monoculture that, in the tradition of colonial rhetoric, "sidelines agencies and cultures on the margins." Thus, if technological relationships between the West and the Global South are indeed influenced by this postcolonial condition, she states, a postcolonial historiography that situates agencies of those at the margins would be vital. In this line, I must say that I have been dealing with some of those different imaginaries of technology in my own work and research for almost twenty years now. In the wake of the popularization of textile technologies as a possible origin of the modern computer, and Ada Lovelace's recognition as the first programmer alongside her popular phrase, the one highly-educated woman who defined Charles Babbage's early prototype as weaving "algebraic patterns just as the Jacquard loom weaves flowers and leaves" came to make

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more sense and began to be considered relevant for some researchers from the Global North, as such validation, inevitably, had to come from the hegemony, and under its own idiosyncratic values. In that sense, a terminology that links to a particular heritage, yet does not simply extract concepts-as-words from Andean worldlings, is needed. *In-Ayllu* (de la Cadena 2015), a term coined by Marisol de la Cadena, follows this syncretic line, yet fails to fully encompass a "pattern thinking" originating from Andean and Australian aboriginal thought, that in turn converges with pattern-based thinking used in contemporary design methodologies, as spouting from Christopher Alexander's "Pattern Language" (Alexander 1977) while acknowledging the woven nature of human-world relations.

In a colloquium in 2013 at Cerisy in France, the popular cyberfeminist Donna Haraway recognized her discomfort when realizing that there was a clear absence of other people's imagining of the world beyond that of Western Europeans, and this was reflected in a predominant absence of thinking from other languages, of knowing and other ways of accessing such knowledge. This was an important lack that should be addressed as, according to her, other words and worldlings "help us reimagine our current urgencies, and perhaps open up a possibility of collaboration and of research" (Haraway 2013).

I grew up in an in-between, in the Andean country of Peru, yet receiving mostly a Westernized education. Now, I live in Europe. And although the Quechua language has been what a portion of my ancestors spoke, I do not speak it. Yet, I am familiar with the Andean way of life, an extremely complex and intricate social system that, although denied by the Peruvian state and some experts, has survived for millennia, still shaping Andean inhabitants' relations between themselves and with the world up to date. This type of social administration became official, according to historians, in Inca times, and it revolves around interconnected nodes where each one is an *Ayllu*, which means *family* - yet it is so much more than just that. As it is mentioned in Marisol de la Cadena´s book, *Earth Beings: ecologies of practice across Andean worlds*, this worldview, in which the living and not are interconnected beyond animism and whose fundamental particle rests in the concept of Ayllu, goes beyond ecocentrism/anthropocentrism binaries, "pointing instead to intra-related practices where earth beings and people shape each other's destiny." In this way, this relational ontology does not only imply connectivism or interrelation, but mutual reshaping through forces both as part of (Nature) but "not only." Positioning myself in the in-between of verbal translation and direct experience without practicing cultural extractivism, I have chosen to combine the Quechua terms *Awasqa* and *Kayku*, "we are pattern/weave" (a *we* that does not include the listener or reader, assumed as an "other" for now) (Callo, 2019) so to continue in the line proposed by de la Cadena in her idea of the "complex we" as composed of an "us" and a "them" occupying and being in a connected time-place while at the same time exhibiting "what makes each (of "us" and "them") be" (de la Cadena 2019, 477-484) in turn linking such concept with Morin´s complex thought as a transdisciplinary paradigm, all partially connected while at the same time exhibiting not difference but divergence, that is, non-oppositional, not binary groupings, but as a branching and re-merging set of knowledges and areas of study, which implies a potential original unity (of the arts, technologies, and sciences). In this framework, disciplines re-emerge relationally as assertions of temporal divergence that are in the historical process of re-uniting again.

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I hereby declare that I am the original author of the work. The editors shall thus not be held responsible for any obligation or legal action that may derive from the work submitted in terms of violation of third parties' rights, whether intellectual property, trade secret or any other right.

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